Customer Churn Prediction using Machine Learning

Project Overview

This project focuses on predicting customer churn using Machine Learning techniques. The dataset used is the Telco Customer Churn dataset, where the goal is to identify customers who are likely to leave a telecom service provider.

The project was completed as part of the BAN644/BAN744 (Applied Machine Learning for Business Analytics with Python) course.

Objective

- Predict customer churn using classification models
- Perform data cleaning and preprocessing
- Conduct Exploratory Data Analysis (EDA)
- Identify important factors influencing customer churn
- Evaluate model performance using classification metrics

Dataset

Dataset: Telco Customer Churn Dataset

The dataset contains customer demographic information, account details, subscribed services, payment methods, and churn status.

Target Variable

- `Churn`
- Yes = Customer left
- No = Customer stayed

Technologies Used

- Python
- Google Colab
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn

Project Workflow

1. Data Loading

- Imported dataset using Pandas
- Checked dataset shape and information

2. Data Cleaning

- Converted `TotalCharges` column to numeric
- Handled missing values
- Removed duplicates

3. Exploratory Data Analysis (EDA)

Performed visual analysis on:

- Churn distribution
- Customer demographics
- Internet services
- Contract types
- Payment methods
- Monthly charges
- Tenure analysis

4. Data Preprocessing

- Label encoding / categorical conversion
- Feature selection
- Train-test split
- Feature scaling using StandardScaler

5. Machine Learning Models

Used classification algorithms such as:

- Random Forest Classifier
- Logistic Regression

6. Model Evaluation

Evaluated models using:

- Accuracy Score
- Classification Report
- Confusion Matrix

Key Insights

- Customers with month-to-month contracts showed higher churn rates.

- Higher monthly charges were associated with increased churn.
- Long-term customers were less likely to churn.
- Certain payment methods and internet services influenced customer retention.

Results

The machine learning model successfully predicted customer churn with good classification performance.

Performance evaluation was done using:

- Accuracy
- Precision
- Recall
- F1-Score

File Structure

```
├── Iffat_Kabir_Kakon_Telco_Customer_Churn.ipynb
├── Telco-Customer-Churn.csv
└── README.md
```

How to Run the Project

1. Clone the repository

```
git clone <your-github-repository-link>
```

1. Open the notebook in Google Colab or Jupyter Notebook

2. Install required libraries

```
pip install pandas numpy matplotlib seaborn scikit-learn
```

1. Run all cells sequentially

Future Improvements

- Hyperparameter tuning
- Trying advanced ML models
- Deployment using Streamlit or Flask

- Building an interactive dashboard

Academic Information

Course: BAN644/BAN744 Topic: Applied Machine Learning for Business Analytics with Python

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